

ABBREVIATED CURRICULUM VITAE

Part A. PERSONAL INFORMATION

CV date

01/12/2025

First name	Saul Ismael		
Family name	Utrera Barrios		
Gender (*)	Male	Birth date (dd/mm/yyyy)	10/04/1992
ID number	60229831F	e-mail	siuba@kt.dtu.dk
Open Researcher and Contributor ID (ORCID)	0000-0003-4604-2305		

A.1. Current position

Position	Postdoc		
Initial date	01/04/2024		
Institution	Technical University of Denmark (DTU)		
Department/Center	Chemical and Biochemical Eng. / Danish Polymer Centre (DPC)		
Country	Denmark	Teleph. number	+45 71504468
Keywords	Polymers, elastomers, rubbers, silicone elastomers, smart materials.		

A.2. Previous professional status

Period	Position/Institution/Country/Interruption cause
2020-2024	Researcher / Institute of Polymer Science and Technology (ICTP), Spanish National Research Council (CSIC) / Spain

A.3. Education

PhD, Licensed, Graduate	University/Country	Year
PhD : Advanced Chemistry	Complutense University of Madrid (UCM) / Spain	2024
MSc : Plastics and Rubbers	Menéndez Pelayo International University (UIMP) / Spain	2019
Eng : Materials Engineering	Simón Bolívar University (USB) / Venezuela	2017

A.4. Predoctoral and postdoctoral stays in national and international institutions

- 2024-Present. **Postdoctoral**: 23 months in DPC, DTU, Kgs. Lyngby, Denmark.
- 2023. **Predoctoral (during PhD)**: 1.5 months in Physical Chemistry and Polymer Science (FYSC) and Brubotics Groups, Vrije Universiteit Brussels (VUB), Brussels, Belgium.
- 2018-2019. **Predoctoral (TFM)**: 12 months in the Polymer Composite Group (PCG), ICTP, CSIC, Madrid, Spain.
- 2017. **Predoctoral (TFG)**: 12 months in Polymers Group II, USB, Caracas, Venezuela.

A.5. Summary of scientific production indicators

Articles 17	Books 1	Book Chapters 6	Patents 2	h-index 13
Cites 959 (Scholar)		Master's Thesis/Internships/Special Courses Supervised 7		

Part B. CV SUMMARY

1. My work focuses on elastomers and elastomer-based composites, applying circular economy principles to extend their lifespan and minimize waste.

I began my research career at USB, where I developed natural rubber composites using *in-situ* generated SiO₂ and ZrO₂ as an alternative to conventional mixing methods. **My Bachelor's Final Work received an Outstanding Mention for exceeding expectations for undergraduate-level work** and was published in Composites Part C.

During my Master's Thesis, I produced novel room-temperature self-healing epoxidized natural rubber nanocomposites, based on hydrogen bonding between the rubber and graphene. This work was published in ACS Omega and **featured on the cover by Editor's Choice**.

My PhD at PCG (ICTP, CSIC), supervised by Prof. M A López-Manchado and Dr. M Hernández Santana, focused on self-healing, recyclable carboxylated nitrile rubber via metal-ligand coordination. The material's unprecedented mechanical properties and industrial scalability attracted Sumitomo Riko (Japan; contract under development) and On Holding AG (Switzerland; contract under negotiation). **I graduated "Cum Laude" and received several**

best thesis awards, including the 2025 Margarita Salas Award from Madrid City Hall (€16K, individual prize).

As a visiting researcher at VUB, I manufactured sustainable, robust XNBR-based soft robots supervised by Profs. G Van Assche, J Brancart, and S Terryn. This led to an **invited publication** for *Materials Horizons* (D1, JIF: 12.2) and a **joint national project granted** (CSIC-VUB, RUBOTICS, PID2022).

In parallel with my PhD, I carried out the lab work for an R&D project with Bridgestone Corporation (Japan), on self-repairing natural rubber (NR) using Diels-Alder and disulfide bonds for tire treads. **This resulted in a granted, licensed patent where I'm a co-inventor.**

Now, I'm a postdoc at DPC, DTU, under Prof. A L Skov. I published a study clarifying misconceptions on polydimethylsiloxane (PDMS) thermal transitions and **co-authored a patent on concatenated ring-based PDMS networks**. I built a consortium for the MSCA-DN proposal RINGFORCE, designed all research plans, drafted the proposal, and acted as co-Project Manager (submitted). **I also co-led a project with Hutchinson SA (France) on ring-based networks.**

I have contributed to 27 Scopus-indexed publications (17 articles, 1 book, 6 book chapters, and 3 proceedings), with an FWCI of 2.634 as first author. ~79% are in the top 25% most cited, ~77% in Q1 journals, and ~42% co-authored with researchers worldwide (Scopus). >90% of my first-author articles are open access, and I am the corresponding author of 5. I have 29 conference contributions (20 international).

2. I have been PI of two competitive projects (250k DKK), and part of 8 projects and 3 industry contracts (1.5M €). I actively engaged in outreach, authoring **12 popular science articles** (e.g., *The Conversation*, *Revista de Plásticos Modernos*). My work has accumulated 1055 saves in Mendeley, 13 scientific news and blogs articles features, and 46 social media discussions (PlumX). **I participated in youth communication activities like *Semana de la Ciencia* and the 4 ESO + Empresa program at CSIC, plus popular dissemination training activities within Videnskab.dk at DTU.** I won an accessit in the science communication contest through photography PHOTOPOL (2019).

I am a member of several national and international societies, with **recurrent invited talks** at the webinars of the Latin American Society of Rubber Technology (SLTC).

3. I have supervised 5 TFM (4 in Spain, 1 in Denmark), 2 internships (Spain), and 2 DTU's special courses (Denmark) (thesis-like programs, where the students gain deep experimental insights before starting larger projects). In 2024, I completed the DTU Teaching Lab (lecturer training) and became an **Invited Teacher** in departmental courses (Polymer Technology; Polymer in Processes and Products). Since 2022, I serve as **Invited Professor** at USB, teaching Elastomers I and II online (>60 h/trimester), sustaining courses otherwise at risk of closure due to the crisis in Venezuela.

I am a verified **reviewer for major publishers** (>30 reviews for Springer Nature, Elsevier, RSC, ACS, MDPI, and Taylor & Francis), including *Nature Comm*, *Adv Funct Mater*, and *ACS Sust Chem Eng*. I also served as a **Guest Editor** for a Special Issue in *Polymers*.

4. My leadership is also evident as a member of the executive organizing committee for the 9th International Conference on Self-Healing Materials (ICSHM) 2024 and the steering committee of the International Network of Rubber Technology (RITC).

My work was also distinguished with best thesis prizes from the Spanish Association of Composite Materials (AEMAC) and *Macromol*, and a diploma as a Top-4 finalist for Best Thesis in Polymers from the Spanish Royal Society of Chemistry's Polymer Specialized Group (GEP). I also received the RSC Materials Horizons 2020 Outstanding Paper Award, one Best Poster and one Best Presentation prize at conferences, and a merit-based master's scholarship.

Part C. RELEVANT MERITS

C.1. Publications

In brackets: Journal Impact Factor (JIF) of the year of publication, Journal category, JIF rank, **JIF Quartile**. After brackets: **CA**: I'm the corresponding author, **X/Y**: author position, **OA**: Open Access, Cites in Google Scholar. **10 selected** in chronological order are:

1. Article. **Utrera-Barrios S**, Yu L, Skov A. "Revisiting the Thermal Transitions of Polydimethylsiloxane (PDMS) Elastomers: Addressing Common Misconceptions with Comprehensive Data". *Macromol Mater Eng*, 2025, 310(8), 2500075. doi: [10.1002/mame.202500075](https://doi.org/10.1002/mame.202500075). (JIF: 4.6, Polymer Science, 25/94, **Q2**). **CA**, 1/3, Cites: 8

2. *Book*. M. Hernández Santana, **S. Utrera-Barrios** (editors). "Rubber Materials: Fundamentals, Sustainability, and Applications". Elsevier 2025, 1st. Ed. ISBN: 9780443289897. doi: [10.1016/C2023-0-50939-5](https://doi.org/10.1016/C2023-0-50939-5).
3. *Article*. **Utrera-Barrios S**, Steenackers N, Terryn S, Hernández Santana M et al. "Unlocking the Potential of Self-Healing and Recyclable Ionic Elastomers for Soft Robotics Applications". *Mater Horiz*, 2024, 11, 708-725. doi: [10.1039/D3MH01312J](https://doi.org/10.1039/D3MH01312J). (JIF: 12.2, Materials Science, Multidisciplinary, 39/439, **D1**). **CA, 1/9, OA, Cites: 15. (Invited article)**
4. *Article*. **Utrera-Barrios S**, Martínez M F, Mas-Giner I, Verdejo R, López-Manchado M A, Hernández Santana M. "New recyclable and self-healing elastomer composites using waste from toner cartridges". *Compos Sci Technol*, 2023, 244, 110292. doi: [10.1016/j.compscitech.2023.110292](https://doi.org/10.1016/j.compscitech.2023.110292). (JIF: 8.3, Materials Science, Composites, 5/35, **Q1**). **CA, 1/6, OA, Cites: 7**
5. *Article*. **Utrera-Barrios S**, Verdugo Manzanares R, Grande A, Verdejo R, López-Manchado M A, Hernández Santana M. "New Insights into the Molecular Structure and Dynamics of a Recyclable and Ionically Crosslinked Carboxylated Nitrile Rubber (XNBR)". *Mater Des*, 2023, 233, 112273. doi: [10.1016/j.matdes.2023.112273](https://doi.org/10.1016/j.matdes.2023.112273). (JIF: 7.6, Materials Science, Multidisciplinary, 80/438, **Q1**). **1/6, OA, Cites: 18**
6. *Article*. **Utrera-Barrios S**, Verdejo R, López-Manchado M A, Hernández Santana M. "Self-healing elastomers: a sustainable solution for automotive applications". *Eur Polym J*, 2023, 190, 112023. doi: [10.1016/j.eurpolymj.2023.112023](https://doi.org/10.1016/j.eurpolymj.2023.112023). (JIF: 5.8, Polymer Science, 9/95, **D1**). **1/4, OA, Cites: 77**
7. *Article*. **Utrera-Barrios S**, Ricciardi O, González S, Verdejo R, López-Manchado M A, Hernández Santana M. "Development of sustainable, mechanically strong, and self-healing bio-thermoplastic elastomers reinforced with alginates". *Polymers*, 2022, 14(21), 4607. doi: [10.3390/polym14214607](https://doi.org/10.3390/polym14214607). (JIF: 5.0, Polymer Science - SCIE, 16/86, **Q1**). **1/6, OA, Cites: 19**
8. *Article*. **Utrera-Barrios S**, Verdugo Manzanares R, Araujo-Morera J, González S, Verdejo R, López-Manchado M A, Hernández Santana M. "Understanding the molecular dynamics of dual crosslinked networks by dielectric spectroscopy". *Polymers*, 2021, 13, 9, 3234. doi: [10.3390/polym13193234](https://doi.org/10.3390/polym13193234). (JIF: 4.967, Polymer Science - SCIE, 16/90, **Q1**). **1/7, OA, Cites: 29. (Most Notable Articles in September-December 2021)**
9. *Article*. **Utrera-Barrios S**, Araujo-Morera J, Pulido de los Reyes L, Verdugo Manzanares R, Verdejo R, López-Manchado M A, Hernández Santana M. "An effective and sustainable approach for achieving self-healing in nitrile rubber". *Eur Polym J*, 2020, 139, 110032. doi: [10.1016/j.eurpolymj.2020.110032](https://doi.org/10.1016/j.eurpolymj.2020.110032). (JIF: 4.598, Polymer Science - SCIE, 15/90, **Q1**). **1/7, Cites: 78.**
10. *Article*. **Utrera-Barrios S**, Verdejo R, López-Manchado M A, Hernández Santana M. "Evolution of self-healing elastomers, from extrinsic to combined intrinsic mechanisms—A review". *Mater Horiz*, 2020, 7(11), 2882-2902. doi: [10.1039/D0MH00535E](https://doi.org/10.1039/D0MH00535E). (JIF: 13.266, Materials Science, Multidisciplinary - SCIE, 26/334, **D1**). **1/4, OA, Cites: 443. (Materials Horizons 2020 Paper Awards) (Top 20 Most Popular Articles 2020) (Top 1% Most Cited Articles 2021)**

C.2. Congress

Five selected as a speaker and first author are:

1. **Utrera-Barrios S**, Woolridge C, Fakhari R, Skov A. "Optimized Silicone Rubbers for Next-Generation Energy Storage". International Conference on Silicone-based Polymers 2025 (ISPO 2025), DTU, Snekkersten, Denmark, 2025. **Oral communication.**
2. **Utrera-Barrios S**, Fakhari R, Woolridge C, Skov A. "Stretch, release, and powered: silicone rubber-based batteries". SPIE Smart Structures + Nondestructive Evaluation 2025: Electroactive Polymer Actuators, Sensors, and Devices (EAPAD 2025), The International Society for Optics and Photonics (SPIE), Vancouver, Canada, 2025. **Demonstration session.**
3. **Utrera-Barrios S**, Mas Giner I, Martínez M F, Verdugo Manzanares R, Verdejo R, López-Manchado M A, Hernández Santana M. "A comparative analysis of ionically crosslinked XNBR composites reinforced with conventional and eco-friendly fillers". International Conference on Self-Healing Materials 2024 (ICSHM 2024), CSIC, Madrid, Spain, 2024. **Oral communication.**

4. **Utrera-Barrios S**, Ricciardi O, Verdejo R, López-Manchado M A, Hernández Santana M. "Self-healable bio-based thermoplastic elastomers reinforced with alginic acid salts". International Elastomer Conference 2022 (IEC 2022), Rubber Division, ACS, Tennessee, United States of America, 2022. **Oral communication**.
5. **Utrera-Barrios S**, Verdugo Manzanares R, Ricciardi O, Verdejo R, López-Manchado M A, Hernández Santana M. "Giving another opportunity to trash: a complete study of recyclability in carboxylated nitrile rubber". Spring Technical Meeting, Rubber Division, ACS, Ohio, United States of America, 2022. **Oral communication**.

C.3. Research projects

I have been the principal investigator (PI) of the following competitive R&D projects. I wrote the proposals and was fully responsible for the scientific laboratory tasks and management:

1. **Title and reference number:** Rubber-based batteries for sustainable energy storage (RubBatts). **Funding body:** DTU Skylab (Denmark). **Participants:** DTU. **Budget:** 150,000.00 DKK. **Duration:** 07/2025 - 01/2026. **PI:** Utrera-Barrios S.
2. **Title and reference number:** Ring Architectures for Multifunctional Silicone Elastomers. **Funding body:** Ministry of Higher Education and Science (Denmark). **Participants:** DTU. **Budget:** 100,000.00 DKK. **Duration:** 06/2025 - 12/2025. **PI:** Espitia Armenta H & Utrera-Barrios S.

I participated as part of the work team and was the main person responsible for the scientific laboratory tasks of the following competitive R&D projects:

1. **Title and reference number:** Rubber band batteries as a green alternative to existing batteries (RUBBA). **Funding body:** Novo Nordisk Foundation (Denmark). **Participants:** DTU. **Budget:** 4,996,632.00 DKK. **Duration:** 01/2025 - 01/2028. **PI:** Skov A, Madsen F, Spinks G, Niordson C.
2. **Title and reference number:** Caucho natural autorreparable: hacia un modelo de economía circular en el sector del transporte (PIE-202060E183). **Funding body:** CSIC (Spain). **Participants:** ICTP. **Budget:** 149,932.31 €. **Duration:** 07/2020 - 01/2024. **PI:** López-Manchado M A.
3. **Title and reference number:** Self-healing nitrile rubber for high abrasion applications (SHENA) (PID2019-107501RB-I00). **Funding body:** Ministry of Science, Innovation and Universities (Spain). **Participants:** ICTP. **Budget:** 96,800.00 €. **Duration:** 06/2020 - 05/2023. **PIs:** Hernández Santana M, Verdejo R.
4. **Title and reference number:** Development of self-healing rubber biocomposites: a sustainable approach – bioHEAL (LINKA20325). **Funding body:** CSIC (Spain). **Participants:** ICTP, Gdańsk University of Technology, University of Wisconsin–Madison, Chalmers University of Technology, Vrije Universiteit Brussel. **Budget:** 23,989.88 €. **Duration:** 01/2021 - 12/2022. **PI:** Hernández Santana M.

C.4. Contracts, technological or transfer merits

Contracts

I participated as part of the work team and was the main person responsible for the laboratory tasks for the following contracts with companies:

1. **Title:** Development of ring PDMS. **Company:** Hutchinson SA (France). **Budget:** 150,000.00 €. **Duration:** 02/2024 - 02/2025. **PI:** Skov A.
2. **Title:** Development of self-healing NR compounds. **Company:** Bridgestone Corporation (Japan). **Budget:** 150,000.00 €. **Duration:** 02/2021 - 10/2022. **PI:** Hernández Santana M.

Transfer merits

1. **European Patent** (Application No. 25171941.5). "Rubber-Based Batteries for Sustainable Energy Storage". **Inventors:** Skov A, Stig Frederiksen N, Yu L, Utrera-Barrios S, Woolridge C D, Fakhari R. **Applicants:** DTU (Denmark). **Status:** Filed, 23/04/2025.
2. **European Patent** (EP4520784A1). "Method for obtaining a self-repairing natural rubber, material thus obtained, and use thereof". **Inventors:** Hernández M, Verdejo R, López Manchado M, Utrera Barrios S. **Applicants:** CSIC (Spain) and Bridgestone Corporation (Japan). **Status:** Licensed, 17/06/2024. Granted, 12/03/2025 (Bulletin 2025/11).